

DATA 2001 Practical Assignment – Task 4

1. Introduction

This report summarises the main findings from Task 1 and Task 3. Task 1 focuses on the derived statistics created from the NSW regional summary dataset and identifies the most interesting patterns in demographics, population growth, housing affordability and business structure.

Task 3 focuses on the POI-based well-resourced scores calculated for SA2 regions in four Greater Sydney SA4 areas. The scores are shown through summary tables, score distributions and map-overlay visualisations to explain how local resource availability differs across the selected regions.

2. Summary of Key Findings from Task 1

Table 1. Summary of the four Task 1 themes and key derived findings.

Themes	Main derived statistics	Key findings
Workforce dependency	Dependency ratio, working-age share, growth gap, replacement proxy	Dependency pressure increased slightly from 2019 to 2024, but NSW still maintained a large working-age population. The youth-to-retirement replacement proxy also suggests a reasonable future workforce pipeline.
Population growth	Natural increase, migration dependence, internal migration offset	NSW's recent population growth was mainly supported by net overseas migration, while natural increase weakened over time. Negative internal migration also offset part of the overseas migration gain.
Housing affordability	Price-income ratio, rent-income ratio, housing stress, disadvantage proxy	High house prices, rent burden and housing stress made affordability a major livability issue in NSW. The results also suggest that housing

		pressure was linked with socioeconomic disadvantage.
Business structure	Industry size, growth and top-three industry concentration	Business activity was concentrated in construction and several large service-based sectors. The top three industries accounted for a substantial share of listed businesses, showing clear industry concentration.

2.1 Workforce dependency and demographic pressure

The workforce dependency statistics show that NSW had a substantial working-age population but also faced gradually increasing dependency pressure. The non-working-age dependency ratio was 54.44 dependents per 100 working-age people in 2024, while the working-age population capacity share was 64.75%. This means that around two thirds of NSW residents were in the working-age group, but the remaining population still created a considerable support load for the working-age population.

The change from 2019 to 2024 is important because it shows that population structure is not static. The working-age population grew, but total population growth was slightly faster, producing a working-age growth gap of -0.61 percentage points. This suggests that children and older residents grew slightly faster than the working-age group, which can increase pressure on labour, services and public support systems.

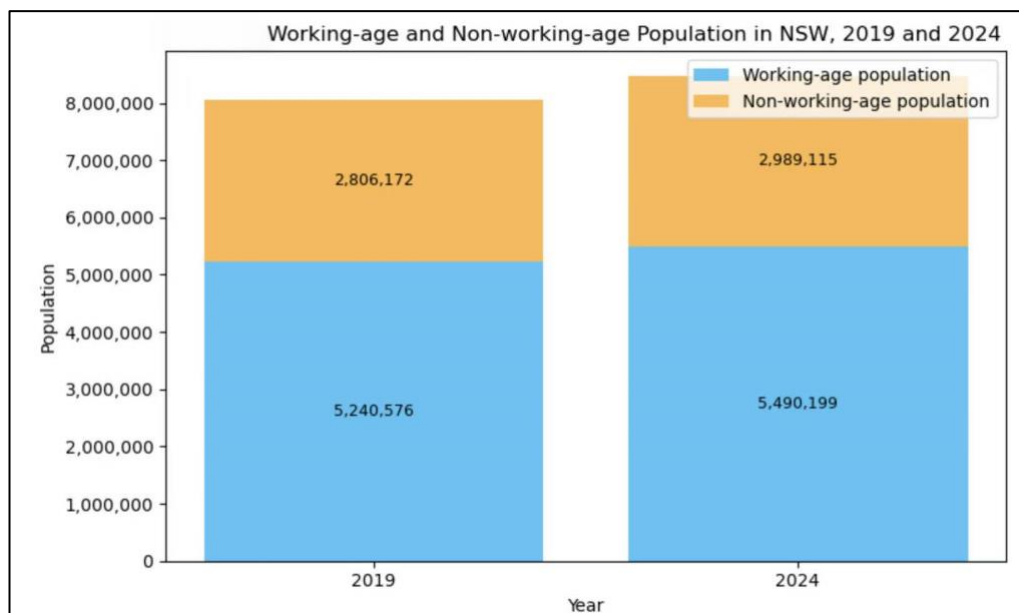


Figure 1. Working-age and non-working-age population in NSW, 2019 and 2024.

According to Figure 1, it shows that both working-age and non-working-age populations increased between 2019 and 2024. However, the relative balance between these groups matters more than the raw totals alone. A slightly higher dependency ratio means that the working-age population may need to support a larger non-working-age group over time.

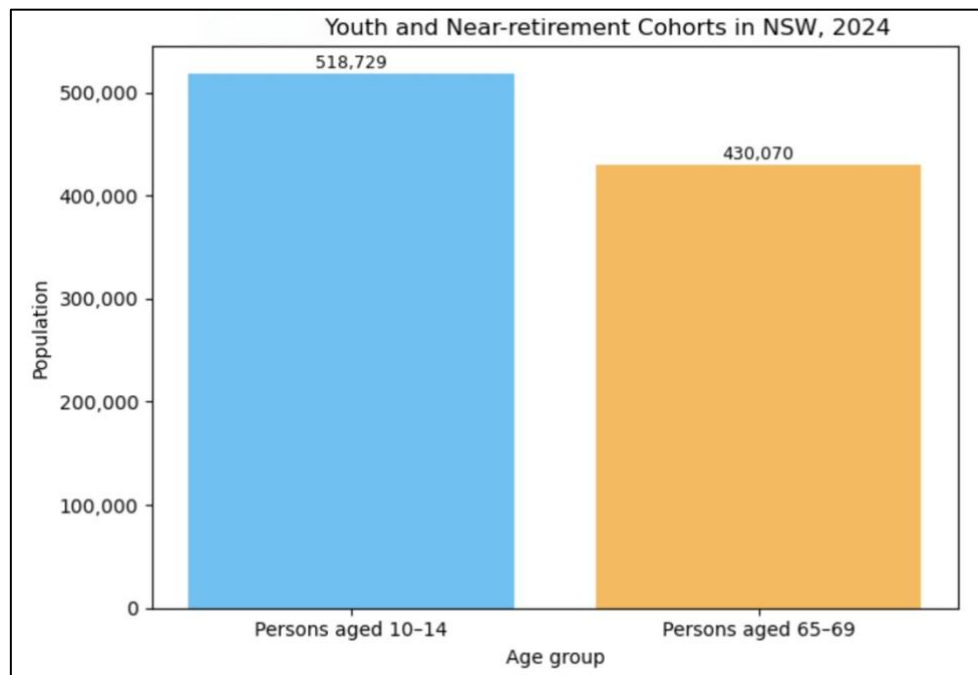


Figure 2. Youth and near-retirement cohorts in NSW, 2024.

Moreover, Figure 2 provides a more positive interpretation of future workforce sustainability. The youth-to-retirement replacement proxy was 120.62%, meaning that the cohort aged 10-14 was larger than the cohort aged 65-69. This suggests that NSW still had a reasonable pipeline of future workers, even though short-term dependency pressure was increasing.

2.2 Population growth drivers and migration balance

The population growth statistics show that NSW population growth in 2023 was strongly shaped by migration. Natural increase remained positive, but it had weakened compared with 2019. The natural increase rate was 3.90 people per 1,000 population, showing that births still exceeded deaths, but the contribution was small relative to the total population. The overseas migration dependence ratio was 101.49%, meaning that net overseas migration was slightly larger than total population growth.

This result is possible because internal migration was negative. NSW gained population through overseas migration but lost population to other Australian states

and territories. The internal migration offset ratio of 19.29% shows that nearly one fifth of the overseas migration gain was offset by internal migration loss. This reveals why total population growth alone is not enough to explain demographic change.

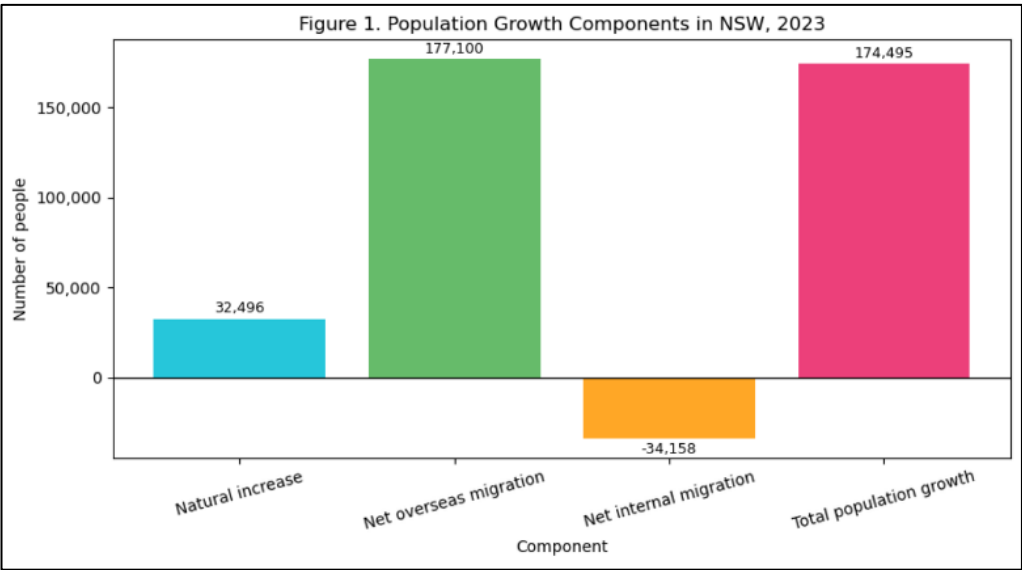


Figure 3. Population growth components in NSW, 2023.

Net overseas migration was the largest positive component of population growth in 2023 based on Figure 3. Natural increase was positive but much smaller, while net internal migration was negative. This supports the finding that NSW's recent growth was driven more by overseas migration than by natural increase alone.

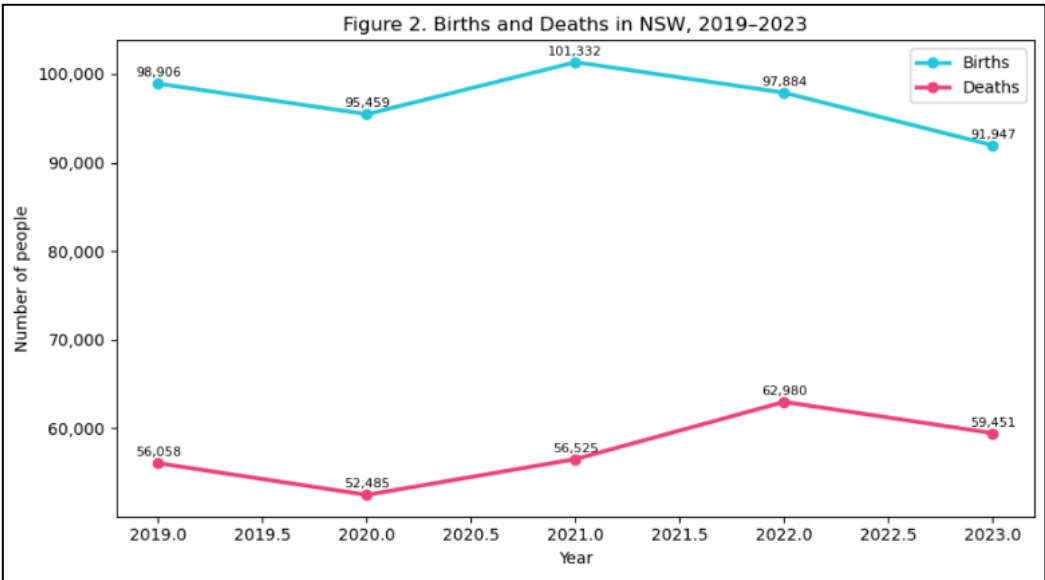


Figure 4. Births and deaths in NSW, 2019-2023.

Figure 4 shows that births remained higher than deaths throughout the period, but the gap narrowed by 2023. Births declined from 2019 to 2023, while deaths increased

overall. This narrowing gap explains why natural increase weakened after 2021 and why migration became more important for total population growth.

2.3 Housing affordability and livability

The housing affordability statistics show that housing affordability was a major livability issue in NSW. The house price-to-income ratio was 12.96 in 2021, meaning that the median established house price was almost 13 times the annual median employee income. The rent-to-income ratio was also high at 37.73%, showing that annual median rent took up a large share of annual median employee income.

The rent-income growth gap was -4.97 percentage points from 2011 to 2021, meaning that median household income grew slightly faster than median weekly rent over this period. However, this does not mean rental pressure was low. The rent-to-income ratio and housing stress figures show that housing costs still placed a substantial burden on residents. The combined housing stress rate was 52.80 percentage points, and the disadvantage-adjusted housing stress proxy was 13.46, indicating that affordability pressure was also connected with socioeconomic disadvantage.

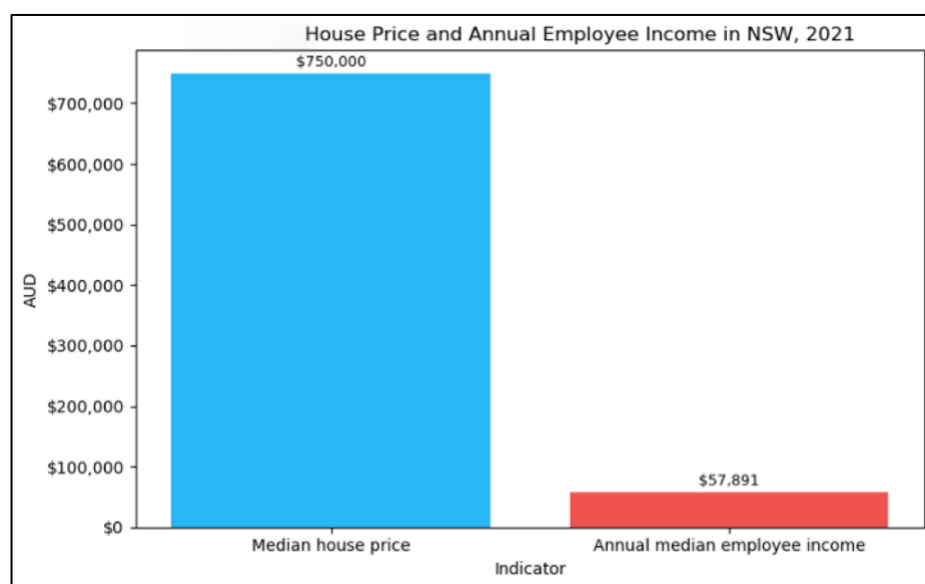


Figure 5. House price and annual employee income in NSW, 2021.

Figure 5 clearly shows the size of the affordability gap. The median house price was \$750,000, while the annual median employee income was \$57,891. This large difference suggests that home ownership was difficult to achieve through ordinary income alone.

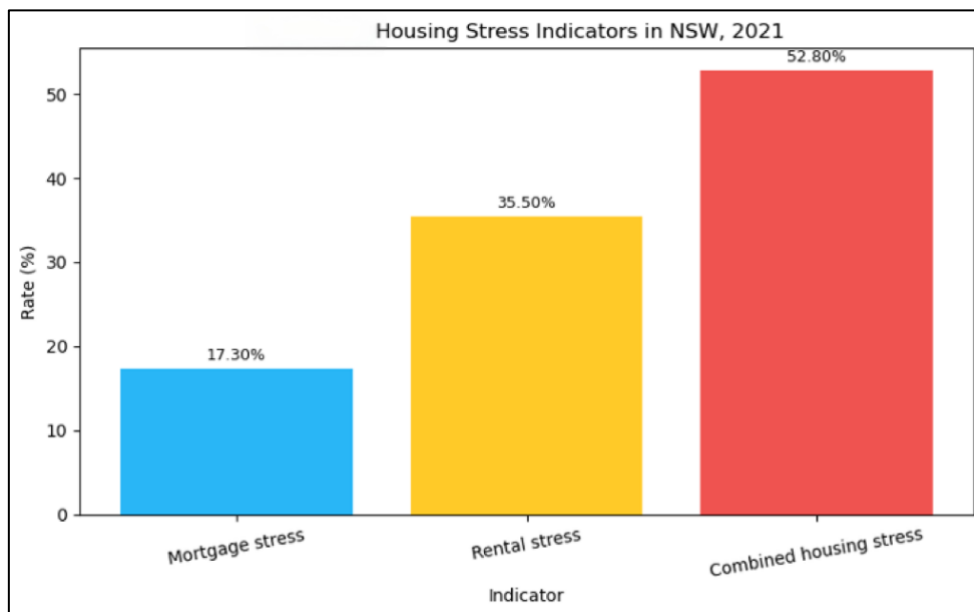


Figure 6. Housing stress indicators in NSW, 2021.

In addition, Figure 6 shows that rental stress was higher than mortgage stress in 2021. Mortgage stress was 17.30%, while rental stress was 35.50%. The combined housing stress rate of 52.80 percentage points indicates that both owner-occupiers and renters faced affordability pressure, but rental stress was the larger contributor.

2.4 Business and industry structure

The business structure statistics show that NSW business activity was concentrated in a small number of large industries. Construction was the largest listed industry in 2024, with 153,172 businesses. It was followed by professional, scientific and technical services, and rental, hiring and real estate services. This indicates that business activity in NSW was strongly led by construction and service-based industries.

The top three industries accounted for 42.4% of listed businesses in 2024, while all other industries together accounted for 57.6%. This means that the NSW business base was not evenly distributed across all industries. Instead, a large share of businesses was concentrated in a few major sectors. This concentration can be important for understanding employment structure, economic resilience and industry-specific growth.

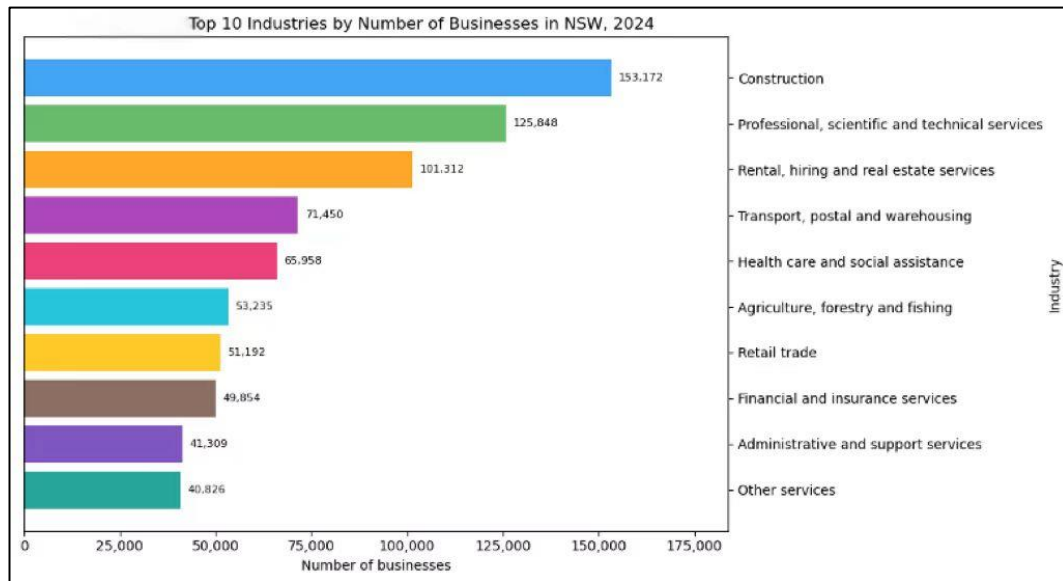


Figure 7. Top 10 industries by number of businesses in NSW, 2024.

Figure 7 shows that construction was clearly the largest industry by number of businesses. Several service-based industries also appeared in the top 10, suggesting that NSW's business structure was shaped by both physical development and service sector activity.

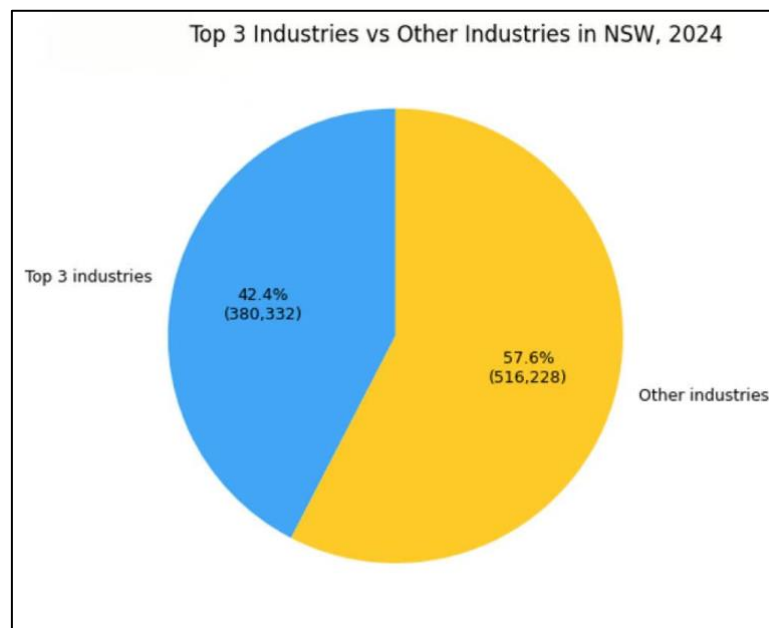


Figure 8. Top three industries versus other industries in NSW, 2024.

Furthermore, Figure 8 reinforces the concentration finding. Although most industries remain outside the top three, the largest three sectors alone accounted for more than two fifths of listed businesses. This makes industry concentration an important part of the overall NSW economic profile.

2.5 Summary

In conclusion, Task 1 shows that NSW faced multiple connected social and economic conditions. Demographic results suggest that the working-age population remained large but carried slightly increasing dependency pressure. Population growth results show that overseas migration was the dominant driver of recent growth, while natural increase weakened. Housing results show that affordability pressure was substantial and linked to both income and disadvantage. Business results show that economic activity was concentrated in major industries, especially construction and service sectors.

The key value of derived statistics is that they make relationships visible. Raw numbers alone show population, income, rent, house prices or business counts, but derived indicators show how these values interact. This makes the analysis more useful for interpreting livability, service demand and regional pressures.

3. POI-based Well-resourced Scores from Task 3

3.1 Scoring method

Task 3 calculated a well-resourced score for SA2 regions in four selected SA4 zones: Sydney - Blacktown, Sydney - North Sydney and Hornsby, Sydney - Parramatta, and Sydney - Northern Beaches. For each SA2, the number of POIs was used as the resource indicator. POI count was selected because areas with more POIs are likely to provide greater access to services, facilities, shops, schools, parks, transport points and local activity centres.

The POI count was standardised within each selected SA4 using a z-score. The z-score measures whether an SA2 has more or fewer POIs than the average SA2 in the same SA4. The sigmoid function was then applied to transform the z-score into a final score between 0 and 1. A score above 0.5 indicates above-average POI-based resource availability within the selected SA4, while a score below 0.5 indicates below-average resource availability.

The Formula used for calculation as following:

$$z_{POI} = \frac{(POI\ count - mean\ POI\ count)}{standard\ deviation\ of\ POI\ count}$$
$$Final\ Score = \frac{1}{(1 + \exp(-z_{POI}))}$$

Therefore, it obtained the following table:

Table 2. Summary of well-resourced scores across selected SA4 regions.

SA4 region	SA2s scored	Highest-scoring SA2	Highest score	Lowest-scoring SA2	Lowest score	Mean score
Sydney - Blacktown	24	Blacktown (East) - Kings Park	0.929	Acacia Gardens	0.211	0.490
Sydney - North Sydney and Hornsby	26	Turramurra	0.966	Artarmon	0.241	0.485
Sydney - Parramatta	34	North Parramatta	0.891	Parramatta - South	0.214	0.491
Sydney - Northern Beaches	19	Bayview - Elanora Heights	0.975	Dee Why (South) - North Curl Curl	0.305	0.479

Table 2 shows that all four selected SA4 regions contained clear internal variation. Each region had a small number of high-scoring SA2 regions and several lower-scoring SA2 regions. The highest individual score was in Sydney - Northern Beaches, where Bayview - Elanora Heights scored 0.975. The lowest score was in Sydney - Blacktown, where Acacia Gardens scored 0.211. However, the mean scores of the four SA4 regions were relatively close, ranging from 0.479 to 0.491. This means that the main variation appeared within SA4 regions rather than between the average scores of the four SA4 regions.

3.2 Sydney – Blacktown

In Sydney - Blacktown, 24 SA2 regions were scored. Blacktown (East) - Kings Park had the highest score of 0.929, while Acacia Gardens had the lowest score of 0.211. The average score was 0.490, close to the midpoint of 0.5. This suggests that Blacktown contained both strongly resourced areas and several areas with lower POI concentration.

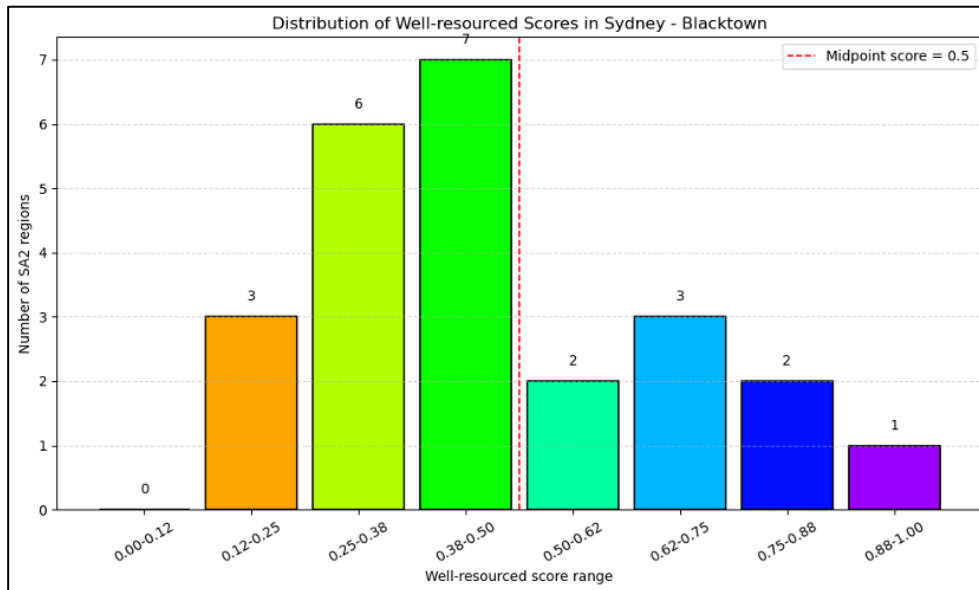


Figure 9. Distribution of well-resourced scores in Sydney - Blacktown.

Figure 9 shows that many Blacktown SA2 regions were clustered in the lower to middle score ranges. Only a few SA2 regions reached very high scores, indicating that high POI-based resource availability was not evenly distributed across the SA4.

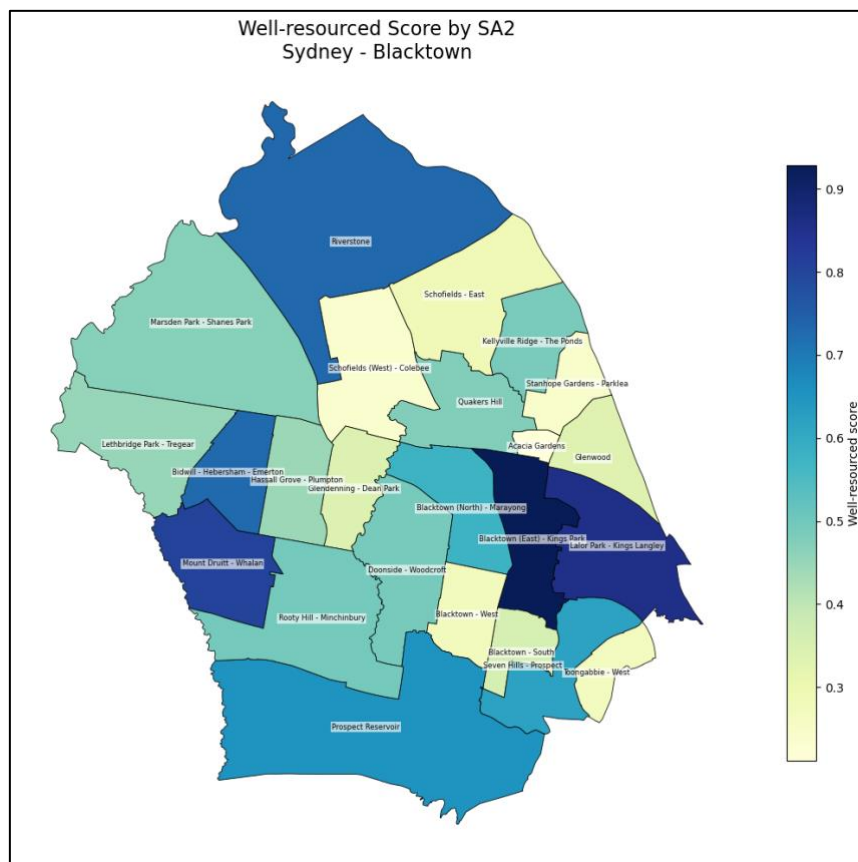


Figure 10. Spatial distribution of well-resourced scores in Sydney - Blacktown.

Moreover, Figure 10 shows that higher-scoring areas were spatially clustered rather than evenly spread. Central and eastern parts of Blacktown tended to have stronger resource scores, while some outer and newer areas had lower scores. This pattern suggests that more established urban areas may have greater POI availability than newer growth areas.

3.3 Sydney - North Sydney and Hornsby

In Sydney - North Sydney and Hornsby, 26 SA2 regions were scored. Turramurra had the highest score of 0.966 and Artarmon had the lowest score of 0.241. The mean score was 0.485 and the median score was 0.436. Although this SA4 includes several high-resource areas, the distribution shows that resource availability still varied noticeably between SA2 regions.

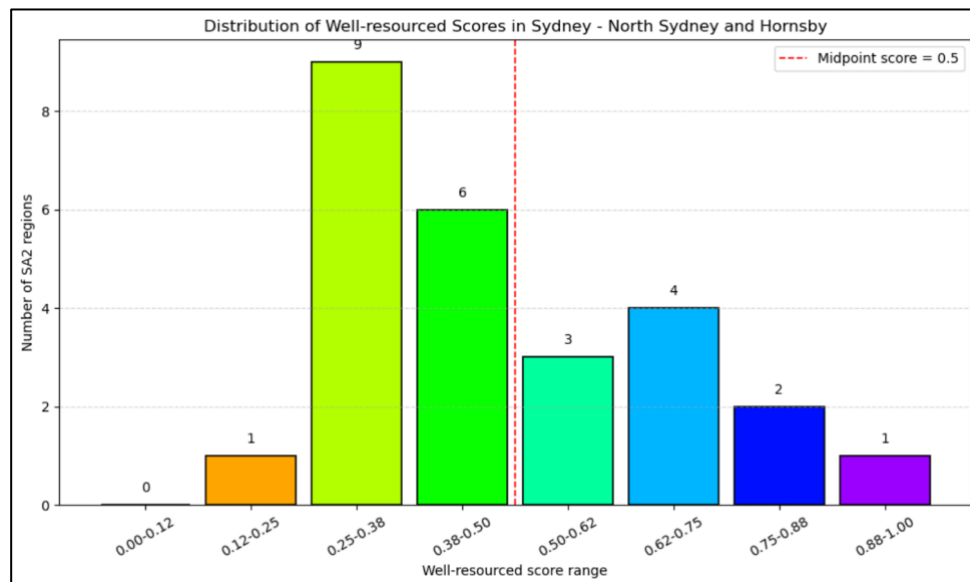


Figure 11. Distribution of well-resourced scores in Sydney - North Sydney and Hornsby.

Figure 11 shows that several SA2 regions were concentrated below or near the 0.5 midpoint, while a smaller group scored much higher. This indicates that even in a generally well-served part of Sydney, local POI-based resources were not evenly distributed.

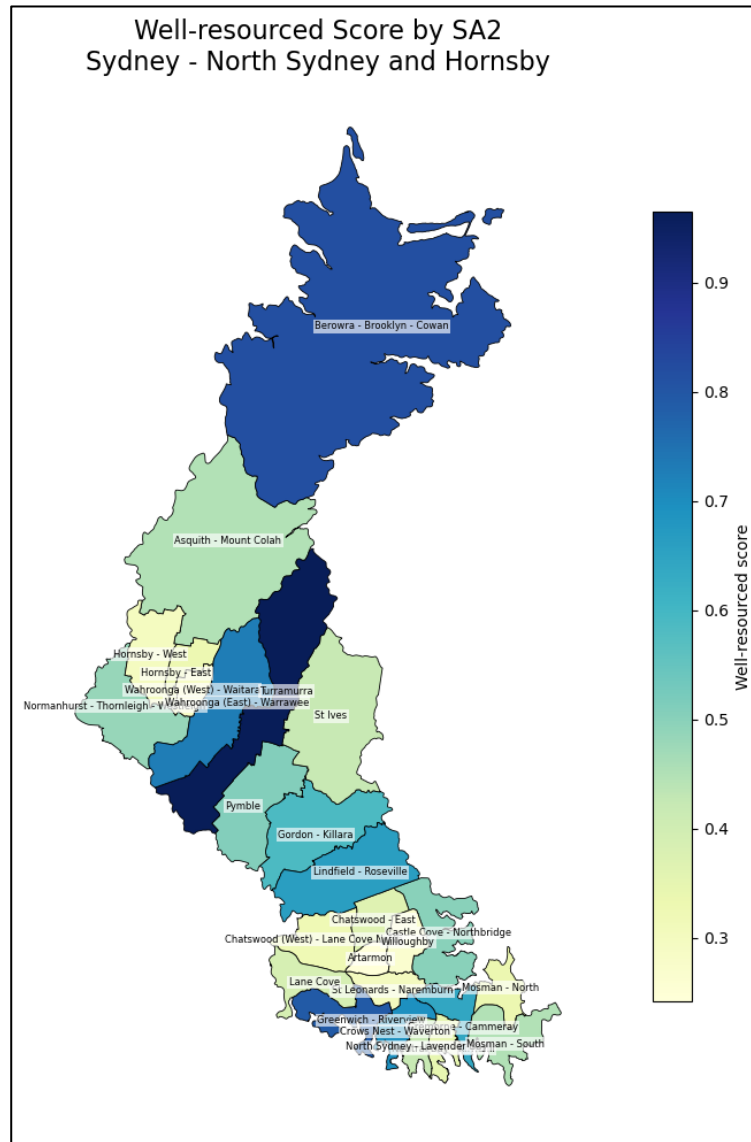


Figure 12. Spatial distribution of well-resourced scores in Sydney - North Sydney and Hornsby.

Additionally, Figure 12 shows the spatial pattern of the scores. Some northern and central SA2 regions had darker colours, indicating higher scores, while other SA2 regions were lighter. This helps identify where high-resource areas appeared geographically rather than only by ranking.

3.4 Sydney – Parramatta

In Sydney - Parramatta, 34 SA2 regions were scored. North Parramatta had the highest score of 0.891 with 177 POIs, while Parramatta - South had the lowest score of 0.214 with only 3 POIs. The average score was 0.491 and the median score was 0.415, suggesting a slightly uneven distribution where a smaller number of high-scoring SA2 regions lifted the average above the median.

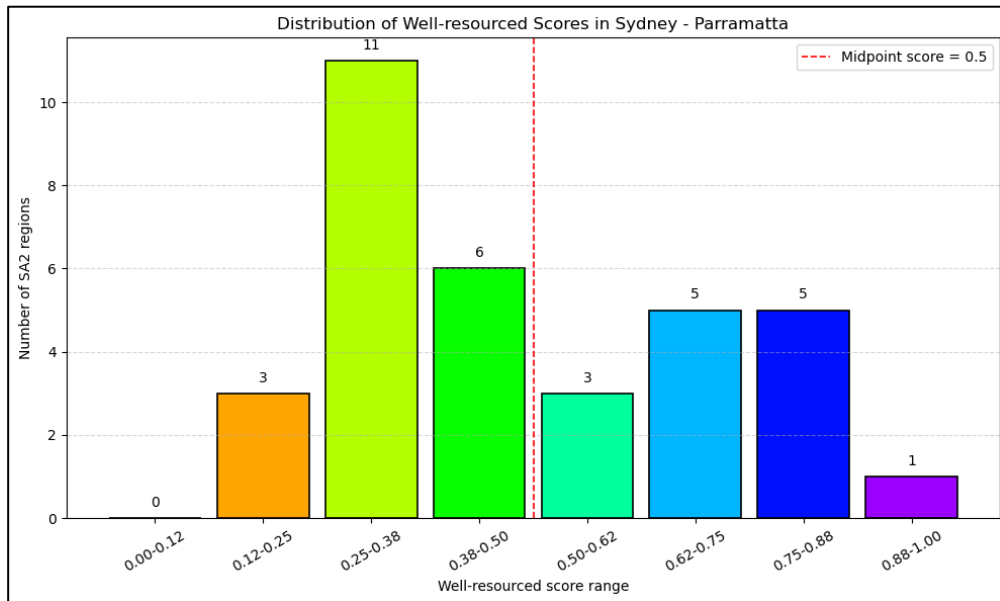


Figure 13. Distribution of well-resourced scores in Sydney - Parramatta.

Figure 13 shows that the largest group of SA2 regions fell within the 0.25-0.38 score range, with 11 SA2 regions in this interval. Another 6 SA2 regions fell within the 0.38-0.50 range. This means many SA2 regions in Parramatta were below the 0.5 midpoint, while a smaller number had high resource scores.

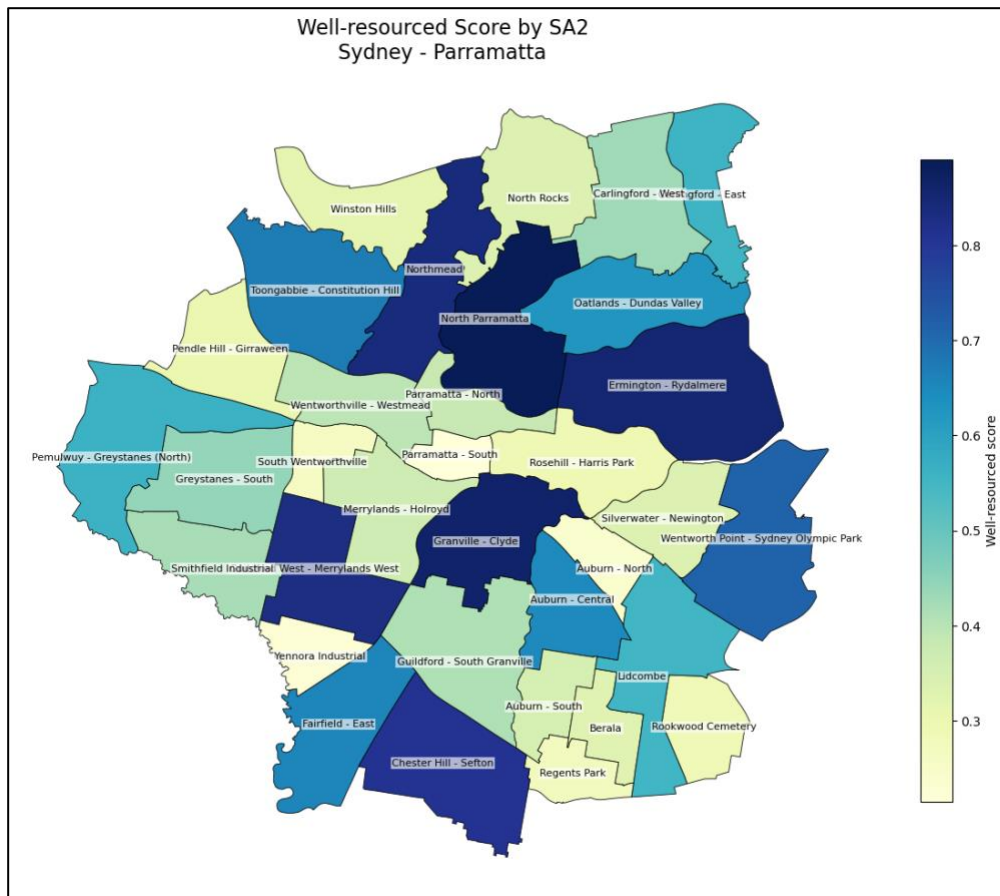


Figure 14. Spatial distribution of well-resourced scores in Sydney - Parramatta.

Furthermore, Figure 14 maps the spatial pattern of well-resourced scores. Darker areas represent higher resource scores and lighter areas represent lower scores. The map shows that resource availability was not evenly distributed across Parramatta. Some central and surrounding SA2 regions had stronger POI-based resource availability, while several others had noticeably lower scores.

3.5 Sydney - Northern Beaches

In Sydney - Northern Beaches, 19 SA2 regions were scored. Bayview - Elanora Heights had the highest score of 0.975 with 684 POIs, while Dee Why (South) - North Curl Curl had the lowest score of 0.305 with 61 POIs. The mean score was 0.479 and the median score was 0.467, both slightly below the midpoint. This indicates that many Northern Beaches SA2 regions were around or below the average-resource level after standardisation and sigmoid transformation.

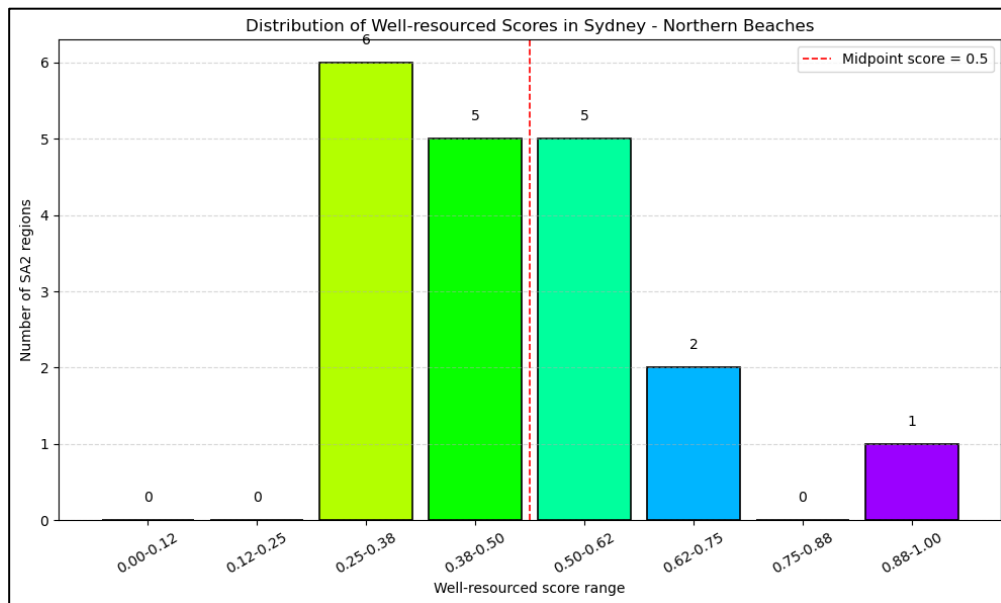


Figure 15. Distribution of well-resourced scores in Sydney - Northern Beaches.

Figure 15 shows that most Northern Beaches SA2 regions were concentrated between 0.25 and 0.62. Only one region was in the highest score range, making Bayview - Elanora Heights a clear high-scoring outlier.

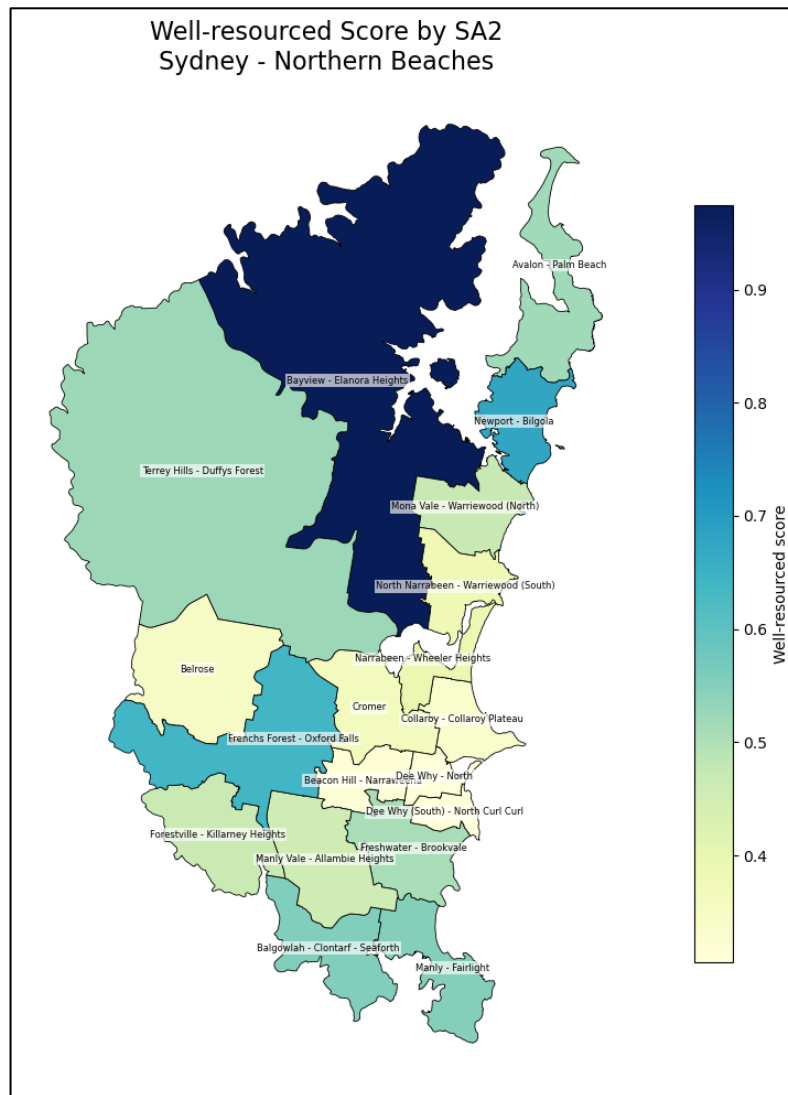


Figure 16. Spatial distribution of well-resourced scores in Sydney - Northern Beaches.

In addition, Figure 16 shows that resource availability varied spatially across Northern Beaches. The highest-scoring area was clearly visible as a darker region, while several smaller coastal and central areas had lower scores. This reinforces the conclusion that POI-based resources were unevenly distributed even within a single SA4.

3.6 Summary

Across all four SA4 regions, the results show that POI-based resource availability was unevenly distributed at the SA2 level. The average scores across the four SA4 regions were similar, but each SA4 contained large differences between its highest- and lowest-scoring SA2 regions. This means that average SA4-level values can hide important local variation.

The map-overlay visualisations are especially useful because they show where high- and low-scoring regions are located geographically. Score distributions show the

spread of scores, but maps reveal whether high-resource areas are clustered in certain parts of each SA4. Overall, Task 3 suggests that local resource availability is spatially uneven and should be examined at the SA2 level rather than only at the larger SA4 scale.

4. Limitations of the Scoring Method

The well-resourced score provides a useful starting point for comparing SA2 regions, but it has several limitations. First, the method is based mainly on POI count, so it measures resource quantity rather than resource quality. A hospital, a school, a park and a small shop may all count as one POI, even though their social and economic importance is very different. This means that high POI counts do not automatically mean high service quality.

Second, all POIs were treated equally. The current method does not distinguish between essential and non-essential services, nor does it account for the size or capacity of facilities. For example, a large hospital and a small private office could both contribute one point. A more advanced scoring method could weight POIs by type or by importance to community wellbeing.

Third, the score does not adjust for population size or service demand. An SA2 region with many POIs may also have a large population using those resources, meaning that resources per person may not actually be high. Conversely, a smaller SA2 with fewer POIs may still feel well-served if its population is small. In the future, the score can be improved by calculating POIs per 1,000 residents.

Fourth, the bounding-box extraction method used during POI collection may introduce spatial uncertainty near SA2 boundaries. Some POIs close to boundaries may be included in the wrong SA2 region, especially where boundaries are irregular. Although duplicate POIs were removed during cleaning, some boundary-related uncertainty may still remain.

Finally, the sigmoid transformation makes the score easy to interpret because it converts z-scores into values between 0 and 1. However, it can also compress very high and very low values. This reduces the impact of extreme outliers, but it may also make the difference between highly resourced areas appear smaller than the raw POI counts suggest. Therefore, the score should be interpreted as a relative indicator, not as a complete measure of accessibility, livability or social equity.

5. Conclusion

This integrated Task 4 report shows that derived statistics and POI-based scoring can reveal patterns that are not obvious from raw data alone. Task 1 showed that NSW faces several connected pressures: demographic dependency, migration-driven population growth, housing affordability stress and industry concentration. These patterns describe the broader social and economic context of NSW.

Then, Task 3 showed that local resource availability varies substantially within selected Greater Sydney SA4 regions. Although the average well-resourced scores across the four SA4 regions were similar, each region contained clear differences between high- and low-scoring SA2 areas. The map-overlay visualisations showed that resources were spatially clustered rather than evenly distributed.

Overall, the analysis suggests that both broad statewide indicators and local spatial indicators are needed to understand regional conditions. Derived statistics help interpret NSW-level demographic, housing and economic pressures, while POI-based maps help identify local differences in resource availability. Together, these methods provide a more complete and practical view of social, economic and spatial patterns across NSW and selected Greater Sydney regions.

6. Acknowledgement of AI use:

ChatGPT was used to support code troubleshooting, grammar checking, and report organisation. All AI-assisted content was reviewed and revised by the group to ensure accuracy and relevance to the assignment requirements.